ECE 6950

Q 3.1 $H_r = U^\dagger H U + i\hbar \frac{\partial U^\dagger}{\partial t} U = e^{i\omega|e\rangle\langle e|t}(\hbar\omega_0|e\rangle\langle e| - d\cdot E)e^{-i\omega|e\rangle\langle e|t} - \hbar\omega|e\rangle\langle e|\,\vec{I}$

a) $= (\hbar\omega_0 - \hbar\omega)|e\rangle\langle e| - U^\dagger d\cdot E U = \begin{bmatrix} \hbar\omega_0 - \hbar\omega & -d\cdot E e^{i\omega t} \\ -d^* E^* e^{-i\omega t} & 0 \end{bmatrix} = \begin{bmatrix} -\hbar\delta & -d\cdot E e^{i\omega t} \\ -d^* E^* e^{-i\omega t} & 0 \end{bmatrix}$

b) $\begin{bmatrix} -\hbar\delta & -d\cdot E e^{i\omega t} \\ -d^* E^* e^{-i\omega t} & 0 \end{bmatrix} \Rightarrow RWA \Rightarrow \begin{bmatrix} -\hbar\delta & -d\cdot E_0 \\ -d^* E_0^* & 0 \end{bmatrix} = -\hbar\begin{bmatrix} \delta & \Omega \\ \Omega^* & 0 \end{bmatrix}$, where Ω = Rabi's frequency

$i\hbar\,\partial_t|\psi\rangle = -\hbar\begin{bmatrix} \delta & \Omega \\ \Omega^* & 0 \end{bmatrix}|\psi\rangle \Rightarrow$ $i\hbar\frac{\partial}{\partial t}c_e = -\hbar\delta c_e - \hbar\Omega c_g$, $i\hbar\frac{\partial}{\partial t}c_g = -\hbar\Omega^* c_e$

Define $\Omega_{eff} = \sqrt{(\delta/2)^2 + |\Omega|^2}$

$\frac{\partial}{\partial t}c_e = i\delta c_e + i\Omega c_g$, $\frac{\partial}{\partial t}c_g = i\Omega^* c_e$ $\Rightarrow$ differentiate $\Rightarrow$ $\partial_t^2 c_e = i\delta\partial_t c_e + i\Omega\partial_t c_g$, $\partial_t^2 c_g = i\partial_t\Omega^* c_e$ $\Rightarrow$ $\partial_t^2 c_e = i\delta\partial_t c_e - \Omega^*\Omega c_e$, $\partial_t^2 c_g = \delta\partial_t c_g - \Omega^*\Omega c_g$

Try ansatz: $c_e = e^{\lambda t}$

$\partial_t^2 c_e = i\delta\partial_t c_e - \Omega^*\Omega c_e \Rightarrow \lambda_e^2 - i\delta\lambda_e + \Omega^*\Omega = 0 \Rightarrow \lambda_e = \frac{i\delta \pm\sqrt{-\delta^2 - 4|\Omega|^2}}{2} = \frac{i\delta}{2} \pm i\Omega_{eff}$

$c_e = C_1 e^{\left(\frac{i\delta}{2} + i\Omega_{eff}\right)t} + C_2 e^{\left(\frac{i\delta}{2} - i\Omega_{eff}\right)t}$ $\quad c_e(0) = 0 \Rightarrow C_1 = -C_2$

$c_e = C_1 e^{\left(\frac{i\delta}{2} + i\Omega_{eff}\right)t} - C_1 e^{\left(\frac{i\delta}{2} - i\Omega_{eff}\right)t} = C_1 e^{\frac{i\delta t}{2}} 2\sin\Omega_{eff} t$

$\frac{\partial}{\partial t}c_g = i\Omega^* c_e \Rightarrow \lambda e^{\lambda t} = i\Omega^* C_1 e^{\frac{i\delta t}{2}} 2\sin\Omega_{eff} t$

Clearly, $c_g = C_3 e^{i\delta t/2} e^{\Omega_{eff} t} + C_4 e^{i\delta t/2} e^{-\Omega_{eff} t}$

$c_g(0) = 1 \Rightarrow c_g = C_3 e^{i\delta t/2} e^{\Omega_{eff} t} + (1 - C_3) e^{i\delta t/2} e^{-\Omega_{eff} t}$

$$c_g = e^{i\delta t/2}\left[\cos(\Omega_{eff} t) + C_3\, i\sin(\Omega_{eff} t)\right]$$

$$\partial_t c_g = e^{i\delta t/2}\left[\left(i\delta/2 + iC_3\Omega_{eff}\right)\cos(\;) + \left(\frac{C_3\delta}{2} - \Omega_{eff}\right)\sin(\;)\right]$$

Comparing coefficients,

$$\cancel{e^{i\delta t/2}}\left[\left(i\delta/2 + iC_3\Omega_{eff}\right)\cos(\ldots) + \left(-\frac{C_3\delta}{2} - \Omega_{eff}\right)\sin(\ldots)\right) = i\Omega^* C_1 \cancel{e^{\frac{i\delta t}{2}}}\, 2\sin\Omega_{eff}t$$

$$C_3 = -\frac{\delta}{2\Omega_{eff}} \qquad C_1 = \left(\frac{\delta^2}{4\Omega_{eff}} - \Omega_{eff}\right)\cdot\frac{1}{i\Omega^*}$$

$$\Omega_{eff}^2 = \frac{\delta^2}{4} - \Omega^*\Omega \Rightarrow C_1 = \left(\frac{\Omega^*\Omega}{\Omega_{eff}}\right)\cdot\frac{1}{i\Omega^*} = -i\frac{\Omega}{\Omega_{eff}} = C_1$$

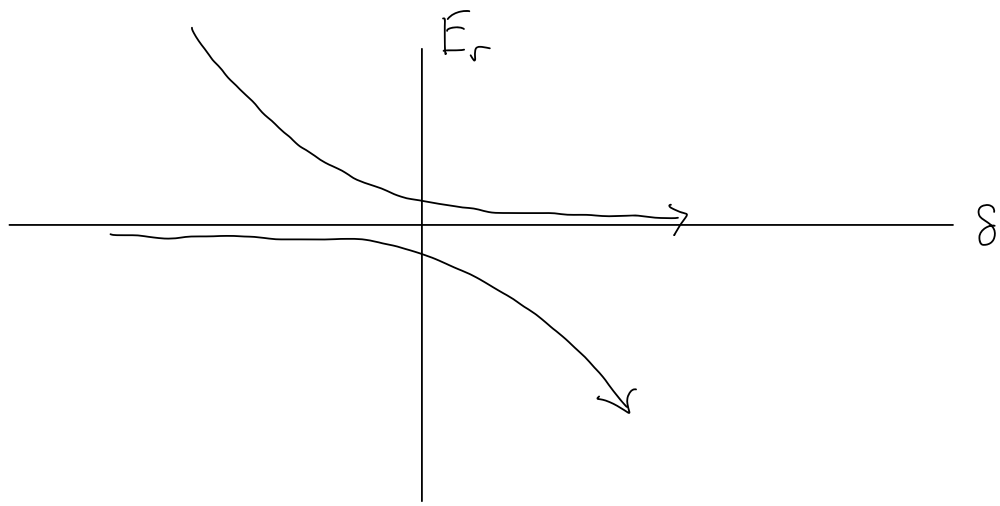
Resulting in,

$$\boxed{C_e = -i\frac{\Omega}{\Omega_{eff}} e^{\frac{i\delta t}{2}} 2\sin\Omega_{eff}t}$$

$$\boxed{C_g = e^{i\delta t/2}\left[\cos(\Omega_{eff}t) - \frac{i\delta}{2\Omega_{eff}}\sin(\Omega_{eff}t)\right]}$$

c)

$$-\hbar\begin{bmatrix}\delta-\lambda & \Omega \\ \Omega^* & -\lambda\end{bmatrix} \Rightarrow \hbar\lambda(\delta-\lambda)+\hbar|\Omega|^2=0 \Rightarrow \lambda = \boxed{-\hbar\left(\frac{\delta \pm \sqrt{\delta^2+4|\Omega|^2}}{2}\right) = E_r}$$



d) Confirm. Both converge to 0 ; $-\infty$

Also visually on graphing

e) $-\hbar\begin{bmatrix}\delta-\lambda & \Omega \\ \Omega^* & -\lambda\end{bmatrix}\begin{bmatrix}a\\ b\end{bmatrix}=\vec{0} \Rightarrow \begin{matrix}\frac{\delta-\sqrt{\delta^2+4|\Omega|^2}}{2}a=-\Omega b \\ \Omega^* a = \frac{\delta+\sqrt{\delta^2+4|\Omega|^2}}{2}b\end{matrix} \Rightarrow \begin{bmatrix}1\\ -\frac{\delta-\sqrt{\delta^2+4|\Omega|^2}}{\Omega 2}\end{bmatrix}$

when $\delta=0$, prop to $\begin{bmatrix}1\\ \sqrt{\frac{-\Omega^*}{\Omega}}\end{bmatrix}$

when $\delta\to\infty$, prop to $\begin{bmatrix}1\\ 0\end{bmatrix}$

$\sigma_x=\begin{bmatrix}0&1\\1&0\end{bmatrix}$

$\sigma_y=\begin{bmatrix}0&-i\\i&0\end{bmatrix}$

$\sigma_z=\begin{bmatrix}1&0\\0&-1\end{bmatrix}$

♡ 3.2) a) $\langle g|\sigma_x|g\rangle=0=\langle e|\sigma_x|e\rangle$

$\langle g|\sigma_x|e\rangle=1=\langle e|\sigma_x|g\rangle$

$\Rightarrow\langle\psi|\sigma_x u_x|\psi\rangle=2\cdot e^{i\phi}\sin\frac{\theta}{2}\cos\frac{\theta}{2}u_x$

$=\cos\phi\sin\theta u_x$

By geometrical/symmetry argument, $\langle\psi|\sigma_y u_y|\psi\rangle=\sin\theta\sin\phi u_y$

$\langle\psi|\sigma_z u_z|\psi\rangle=\cos\theta\cdot 1=\cos\theta$

Collecting, $\langle\sigma\rangle=\sin\theta\cos\phi u_x+\sin\theta\sin\phi u_y+\cos\theta u_z$

$\overset{\sigma_x}{(\pm1,0,0)}, \overset{\sigma_y}{(0,\pm1,0)}, \overset{\sigma_z}{(0,0,\pm1)}$

b) $H_n|\psi\rangle=i\hbar\frac{d}{dt}|\psi\rangle \Rightarrow\begin{bmatrix}0&\hbar\Omega\\ \hbar\Omega^*&0\end{bmatrix}|\psi\rangle=i\hbar\frac{d}{dt}|\psi\rangle$

This matches the initial config above, but $\delta = 0$.

$c_g = 1$, $c_e = 0$ @ $t = 0$

$\Rightarrow$ At $\delta = 0$, we already said $|\psi\rangle = \begin{bmatrix} 1 \\ \sqrt{\frac{\Omega^*}{\Omega}} \end{bmatrix}$ from 1e

Prior result:
$$c_e = -i \frac{\Omega}{\Omega_{eff}} 2 \sin \Omega_{eff} t$$
$$c_g = \cos(\Omega_{eff} t)$$
$$\Rightarrow |\psi\rangle = \begin{bmatrix} -i \frac{\Omega}{\Omega_{eff}} 2 \sin \Omega_{eff} t \\ \cos(\Omega_{eff} t) \end{bmatrix}$$

$\psi(0) = |g\rangle$ as desired

c) For Pauli matrices, $e^{ia(\hat{n} \cdot \vec{\sigma})} = I \cos a + i(\hat{n} \cdot \vec{\sigma}) \sin a$

$$e^{-i t/\hbar \cdot \hbar \begin{bmatrix} 0 & \Omega e^{-i\phi} \\ \Omega e^{i\phi} & 0 \end{bmatrix}} = \vec{I} \cos(\Omega t) - i \begin{bmatrix} 0 & e^{-i\phi} \\ e^{i\phi} & 0 \end{bmatrix} \sin \Omega t$$

$$= \begin{bmatrix} \cos(\frac{\theta}{2}) & -ie^{-i\phi} \sin(\frac{\theta}{2}) \\ -ie^{i\phi} \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{bmatrix} = \hat{U}_\varphi(\theta)$$

The phase ϕ of Ω_r rotates it about the x-y plane.

d) $H_r = -\hbar \begin{bmatrix} 0 & \Omega e^{-i\phi} \\ \Omega e^{i\phi} & 0 \end{bmatrix} = -\hbar\Omega(\ldots)$

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $\quad a - ib = e^{-i\phi}$

$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ $\quad a + ib = e^{i\phi}$

$$a = \frac{e^{-i\phi} + e^{i\phi}}{2} = \cosh(i\phi)$$
$$b = \frac{e^{-i\phi} - e^{i\phi}}{2i} = i \sinh(i\phi)$$

$$\Rightarrow H_r = -\hbar\Omega \left(\sigma_x \cosh(i\phi) + \sigma_z i \sinh(i\phi) \right)$$

In Heisenberg $\frac{\partial \hat{O}}{\partial t} = \frac{i}{\hbar}[\hat{H}, \hat{O}] \Rightarrow \frac{\partial \hat{\sigma}_x}{\partial t} = \frac{i}{\hbar}\Big[\mathrm{Re}\,\Omega \cancel{[\sigma_x \sigma_x]}^{\,0} \cosh \ldots$

$-\hbar\Omega\, i \sinh(i\phi)[\sigma_z, \sigma_x]\Big)$

$$\frac{\partial \hat{\sigma}_x}{\partial t} = -2\Omega \sinh(i\phi)\hat{\sigma}_y = -2\,\mathrm{Re}(\Omega)\, y$$

$$\Omega \times \hat{\sigma} = \begin{bmatrix} -2\mathrm{Re}\,\Omega \\ -2\,\mathrm{Im}(\Omega^*) \\ -\delta \end{bmatrix} \times \begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \end{bmatrix} = \text{Above by Wolfram.}$$

e) Recall $U_\phi(\theta) = \begin{bmatrix} \cos(\theta/2) & -ie^{-i\phi}\sin(\frac{\theta}{2}) \\ -ie^{i\phi}\sin\frac{\theta}{2} & \cos(\theta/2) \end{bmatrix}$

i. $U_0(\pi/2) = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix} \Rightarrow U_0(\frac{\pi}{2})\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} -i \\ 1 \end{bmatrix}$

ii. It then precesses about $\hat{z} \Rightarrow$ Apply for $\tau \Rightarrow \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -ie^{i\delta\tau} \end{bmatrix}$

iii. Apply $U_\phi(\frac{\pi}{2}) = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -ie^{-i\phi} \\ -ie^{i\phi} & 1 \end{bmatrix} \Rightarrow \frac{1}{2}\begin{bmatrix} 1 & -ie^{-i\phi} \\ -ie^{i\phi} & 1 \end{bmatrix}\begin{bmatrix} -ie^{i\delta\tau} \\ 1 \end{bmatrix}$

$$\frac{1}{2}\begin{bmatrix} 1 & -ie^{-i\phi} \\ -ie^{i\phi} & 1 \end{bmatrix}\begin{bmatrix} -ie^{i\delta\tau} \\ 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -ie^{-i\phi} - ie^{i\delta\tau} \\ 1 - e^{i(\delta\tau - \phi)} \end{bmatrix} = |c\rangle$$

$$|c_e|^2 = \left(\frac{-i}{2}\right)\left(\frac{i}{2}\right)\left(e^{-i\phi} + e^{i\delta\tau}\right)\left(e^{i\phi} + e^{-i\delta\tau}\right) = \frac{1}{4}\Big(2 + \underbrace{e^{i(\phi+\delta\tau)} + e^{-i(\phi+\delta\tau)}}_{2\cos(\phi+\delta\tau)}\Big)$$

$$\therefore |c_e|^2 = \frac{1}{4}(2 + 2\cos(\phi + \delta\tau)) = \boxed{\frac{1}{2}(1 + \cos(\phi + \delta\tau))} \checkmark$$

f) We compute $\langle |c_e|^2 \rangle = \int_{-\infty}^{\infty} p(\delta)\,|c_e(\delta)|\,d\delta$

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{1}{2}\left(\frac{\delta - \delta_0}{\sigma}\right)^2\right) \cdot \frac{1}{2}(1 + \cos(\phi + \delta\tau))\, d\delta$$

$$= \cancel{\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{1}{2}\left(\frac{\delta-\delta_0}{\sigma}\right)^2\right)}^{\,1} \cdot \frac{1}{2}\, d\delta + \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{1}{2}\left(\frac{\delta-\delta_0}{\sigma}\right)^2\right) \frac{1}{2}\cos(\phi + \delta\tau)\, d\delta$$

$$= \frac{1}{2} + \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{1}{2}\left(\frac{\delta-\delta_0}{\sigma}\right)^2\right) \frac{1}{2}\cos(\phi + \delta\tau)\, d\delta$$

$$= \frac{1}{2} + \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{1}{2}\left(\frac{\delta-\delta_0}{\sigma}\right)^2\right) \left[\frac{1}{2}\cos(\phi)\cos(\delta\tau) - \cancel{\frac{1}{2}\sin\phi \sin(\delta\tau)}^{\,0 \text{ by symmetry}}\right] d\delta$$

$$= \frac{1}{2} + \frac{\cos(\phi)}{2} \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{1}{2}\left(\frac{\delta-\delta_0}{\sigma}\right)^2\right) \cos(\delta\tau)\, d\delta$$

To solve convert $\cos(\delta\tau)$ to $e^{i\delta\tau}$

$$\Rightarrow \; = \exp\left(-\frac{\tau^2\sigma^2}{2}\right)\cos(\delta_0\tau)$$

$$\Rightarrow \quad \boxed{\langle |c_e|^2 \rangle = \frac{1}{2}\left[1 + e^{-\frac{\tau^2\sigma^2}{2}} \cos(\phi)\cos(\delta_0\tau)\right]}$$

So the $\langle |c_e|^2 \rangle$ decays with τ by the $e^{-\frac{\tau^2\sigma^2}{2}}$ term. It decays exponentially until its baseline probability of 50%. This is typical decoherence.

Q3.3 a) $\hat{U} = \begin{bmatrix} e^{i\lambda_g t} & 0 & 0 \\ 0 & e^{i\lambda_f t} & 0 \\ 0 & 0 & e^{i\lambda_e t} \end{bmatrix} \Rightarrow \frac{\partial U}{\partial t} = \sum_j i\lambda_j e^{i\lambda_j t} |j\rangle\langle j|$

$$i\hbar \frac{\partial U}{\partial t} U^\dagger = -\hbar \begin{bmatrix} \lambda_g e^{i\lambda_g t} & 0 & 0 \\ 0 & \lambda_f e^{i\lambda_f t} & 0 \\ 0 & 0 & \lambda_e e^{i\lambda_e t} \end{bmatrix} \begin{bmatrix} e^{-i\lambda_g^* t} & 0 & 0 \\ 0 & e^{-i\lambda_f^* t} & 0 \\ 0 & 0 & e^{-i\lambda_e^* t} \end{bmatrix} = -\hbar \sum_j \lambda_j |j\rangle\langle j|$$

To match the desired Hamiltonian: $\lambda_g = 0$.

$$U^\dagger H_0 U = \left(\sum_j e^{-i\lambda_j^* t} |j\rangle\langle j|\right) \hbar\omega_f |f\rangle\langle f| + \hbar\omega_e |e\rangle\langle e| \left(\sum_j e^{i\lambda_j t} |j\rangle\langle j|\right)$$

$$= \hbar\omega_f e^{i(\lambda_f - \lambda_f^*)t} |f\rangle\langle f| + \hbar\omega_e e^{i(\lambda_e - \lambda_e^*)t} |e\rangle\langle e| \Rightarrow \hbar\omega_e e^{i(\lambda_e - \lambda_e^*)t} - \hbar\lambda_e = \delta_p$$

True $\forall t \Rightarrow \lambda_e$ is real $\Rightarrow \hbar\omega_e - \hbar\lambda_e = \delta_p \Rightarrow \lambda_e = \omega_e - \frac{\delta_p}{\hbar}$

Similarly, $\hbar\omega_f e^{i(\lambda_f - \lambda_f^*)t} - \hbar\lambda_f = \delta_p - \delta_c \Rightarrow \lambda_f = \omega_f - \frac{\delta_p - \delta_c}{\hbar}$

No need to examine off diagonal contributions

$$\lambda_c = 0 \qquad \lambda_f = \omega_f - \frac{\delta_p - \delta_c}{\hbar} \qquad \lambda_e = \omega_e - \frac{\delta_p}{\hbar}$$

b) $i\hbar \frac{\partial}{\partial t} |\psi\rangle = H|\psi\rangle$. Note $\delta_p = \delta_c = \delta$

$$i\hbar \begin{bmatrix} \dot{c}_g \\ \dot{c}_f \\ \dot{c}_e \end{bmatrix} = \hbar \begin{bmatrix} 0 & 0 & \Omega_p \\ 0 & \delta_p - \delta_c & \Omega_c \\ \Omega_p & \Omega_c & \delta_p \end{bmatrix} \begin{bmatrix} c_g \\ c_f \\ c_e \end{bmatrix} \Rightarrow i \begin{bmatrix} \dot{c}_g \\ \dot{c}_f \\ 0 \end{bmatrix} = \begin{bmatrix} \Omega_p c_e \\ \Omega_c c_e + (\cancel{\delta_p - \delta_c})^{0} c_f \\ \Omega_p c_g + \Omega_c c_f + \delta_p c_e \end{bmatrix}$$

$$\Rightarrow c_e = -\frac{\Omega_p c_g + \Omega_c c_f}{\delta} \Rightarrow \begin{aligned} \partial_t c_g &= -\frac{\Omega_p}{\delta}(\Omega_p c_g + \Omega_c c_f) \\ \partial_t c_f &= -\frac{\Omega_c}{\delta}(\Omega_p c_g + \Omega_c c_f) \end{aligned}$$

$$\Rightarrow \overset{\text{Ansatz}}{e^{\lambda t}} \Rightarrow V \cdot \begin{bmatrix} c_g \\ c_f \end{bmatrix} = -\underbrace{\frac{\Omega_p \Omega_c}{\delta}}_{\Omega_{eff}} \begin{bmatrix} \frac{\Omega_p}{\Omega_c} & 1 \\ 1 & \frac{\Omega_c}{\Omega_p} \end{bmatrix} \begin{bmatrix} c_g \\ c_f \end{bmatrix}$$

Ω_{eff} as desired as coupling.

c) $\hat{H} = \hbar \begin{bmatrix} 0 & 0 & \Omega_p \\ 0 & \delta_p - \delta_c & \Omega_c \\ \Omega_p & \Omega_c & \delta_p \end{bmatrix} = \hbar \begin{bmatrix} 0 & 0 & \Omega_p \\ 0 & 0 & \Omega_c \\ \Omega_p & \Omega_c & \delta \end{bmatrix} \Rightarrow \text{Eig} \begin{bmatrix} 0 & 0 & \Omega_p \\ 0 & 0 & \Omega_c \\ \Omega_p & \Omega_c & \delta \end{bmatrix}$

$$\begin{vmatrix} -\lambda & 0 & \Omega_p \\ 0 & -\lambda & \Omega_c \\ \Omega_p & \Omega_c & \delta - \lambda \end{vmatrix} = 0 \Rightarrow -\lambda\left[-\lambda(\delta-\lambda) - \Omega_c^2\right] + \Omega_p^2 \lambda = 0 \Rightarrow \lambda = 0 \text{ or } -\lambda(\delta - \lambda) = \Omega_c^2 + \Omega_p^2$$

$$\Rightarrow \lambda = 0, \quad \frac{\delta \pm \sqrt{\delta^2 + 4(\Omega_p^2 + \Omega_c^2)}}{2} \Rightarrow \vec{V} = \begin{bmatrix} -\Omega_c/\Omega_p \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} \Omega_c/\lambda_- \\ \Omega_p/\lambda_- \\ 1 \end{bmatrix}, \begin{bmatrix} \Omega_c/\lambda_+ \\ \Omega_p/\lambda_+ \\ 1 \end{bmatrix}$$

Let λ_+, λ_- be this eigenvalue pair

d) Clearly, $c_e|e\rangle$ is unaffected.

So $\langle e|\hat{H}_r|e\rangle = \begin{bmatrix} x \\ x \\ \delta \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \delta \Rightarrow$ Directly impacted by resonance phase difference